Hyesu Jang

Robotics Researcher

hyesu.official@gmail.com Seoul, KR

Summary

Robotics Researcher at Seadronix Inc., focusing on Sensor Fusion and SLAM algorithm development for autonomous vessel systems. Ph.D. in Mechanical Engineering from Seoul National University with expertise in Robust Radar SLAM for Maritime Environments.

Education

Seoul National University (SNU), Mechanical Engineering — Seoul, Korea
Sept 2021 – Aug 2024
- Robust Perception and Mobile Robotics Lab, advised by Prof. Ayoung Kim
- Thesis: Robust Radar SLAM for Maritime Environment

Korea Advanced Institute of Science and Technology (KAIST), Civil and Environmental Engineering — Daejeon, Korea
Sept 2020 – Aug 2021
- Advised by Prof. Ayoung Kim

Korea Advanced Institute of Science and Technology (KAIST), Civil and Environmental Engineering — Daejeon, Korea
Sept 2018 – Aug 2020
- Advised by Prof. Ayoung Kim
- Thesis: Opti-Acoustic Underwater SLAM Using Style-transferred Multi-modal Matching

Korea Advanced Institute of Science and Technology (KAIST), Civil and Environmental Engineering — Daejeon, Korea
Mar 2014 – Aug 2018

Warsaw University of Technology (WUT), Exchange Student — Warsaw, Poland
Mar 2017 – June 2017

Busan Science High School, Physics, Electronic Engineering — Busan, Korea
Mar 2012 – Feb 2014

Experience

Seadronix Inc., Robotics Researcher — Seoul, Korea
Jan 2025 – present
1 year 11 months
Sensor Fusion and SLAM algorithm development for autonomous vessel systems utilizing X-band Radar, W-band Radar, and LiDAR.

Seoul National University, Post-Doctoral Researcher — Seoul, Korea
Sept 2024 – Dec 2024
4 months
Institute of Advanced Machines and Design.

Awards

Outstanding Young Researcher Award 2026

Best Conference Paper Award – Finalist 2025

Best Paper Award on Robot Perception – Finalist 2025

Excellent Award (Runner-up) 2017

Grand Prize 2016

Encouragement Award (3rd Prize) 2016

Patents

Radar placement and algorithm for ego-motion estimation
Registered, No.10-2587292

Electronic device for estimating underwater path with acoustic and optical matching
Registered, No.10-2678475

A method for performing place recognition using radar image
Filed, No.10-2025-0136094

Apparatus and method for determining odometry of mobile object
Filed, No.10-2026-0026295

Projects

High-precision sensor system and software for autonomous return of small and medium-sized vessels in coastal areas 2025
Seadronix Inc.

Development of multipurpose offshore floating infrastructure construction technology 2023-2024
Seoul National University

Maritime Radar Odometry Estimation 2023
Seoul National University

Non-optical sensor-based autonomous driving for extreme environments 2021-2024
Seoul National University

Construction site information collection and analysis technology 2021-2024
Seoul National University

Automated urban analysis techniques using 3D sensor data 2019
KAIST

Navigation Technology and Safety Device Design for the Visually Impaired 2017
KAIST Undergraduate Research Program

Weather/Season Invariant Place Recognition using Image Processing 2016
KAIST Undergraduate Research Program

Workshop Organization

Workshop on Radar in Robotics: Resilience from Signal to Navigation 2024

Professional Service

Reviewer
IEEE Transactions on Robotics (T-RO) IEEE Robotics and Automation Letters (RA-L) IEEE International Conference on Robotics and Automation (ICRA) IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) International Conference on Ubiquitous Robots (UR) Journal of Korea Robotics Society (JKROS)

Teaching Assistant
[CE352] Signal and System for Construction IT, KAIST (2018) [M3228.000300] Sensor-Based Spatial Intelligence, SNU (2022) [M2794.004200] Mechanical System Design Project, SNU (2022)

Page/Server Administrator
MOANA Dataset MulRan Dataset SNU RPM Lab website

Skills

Programming

Sensor Fusion & SLAM

Robotics

Tools